

IMPERIAL

Experimental and Computational Investigation of Small-Scale Turbopump Performance with a Novel Barske-Type Impeller



Student: Martin Leung
CID: 02217557
Academic Supervisor: Dr Aaron Knoll
Second Marker: Dr Roderick Lubbock
Department: Department of Aeronautics
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Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.

Abstract

Your abstract goes here. Summarise the key motivations, methodology, and findings of your turbopump FYP.

Acknowledgements

I would like to thank...

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Nomenclature

A	Area (m^2)
H	Pump Head (m)
N	Rotational Speed (RPM)
P	Power (W)
Q	Volumetric Flow Rate (m^3/s)
η	Efficiency ($-$)
ρ	Density (kg/m^3)
ω	Angular Velocity (rad/s)

Abbreviations

CFD	Computational Fluid Dynamics
FYP	Final Year Project
NPSH _a	Net Positive Suction Head Available
NPSH _r	Net Positive Suction Head Required

1 Introduction

Introduction text goes here.

2 Pump Design Theory

Pump sizing starts with determining the required flow rate and head. As the pump for this report serves to provide consistent fuel flow into the rocket injector, the flow rate and head are predetermined.

The rocket engine requires a delivery pressure of 20 bar and a mass flow rate of 0.3

3 Conclusion

Conclusion text goes here.

References

A First Appendix Title

Appendix content goes here (e.g., calibration curves, large tables, or code).